



Beyond Correctness: A Stage-Aware Framework for Decoding Student Problem-Solving Processes from Handwriting Trajectories

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Abstract. Process-oriented assessment is essential for adaptive learning, yet most educational systems still rely on outcome-based signals like correctness, offering limited insight into how students regulate their problem-solving. This limits our ability to diagnose breakdowns in metacognitive monitoring and strategic planning, and provide targeted support. To bridge this gap, we introduce a stage-aware assessment framework that transforms raw handwriting trajectories from digital pens into a multi-level behavioral representation. Leveraging Multimodal Large Language Models (MLLMs) as an interpretive layer, the framework maps low-level pen strokes to functional cognitive operations and situates them within a theory-grounded problem-solving process. Analyzing real-world exam instances from a semester-long dataset, we derive stage-aware features that characterize regulatory events, question-level dynamics, and stable behavioral profiles. Our analysis reveals distinct learner profiles, including students who engage in structured planning and revision, and those who proceed impulsively with minimal intermediate reasoning. We further uncover a strategy-style mismatch: some students invest substantial effort without matching gains, underscoring the limits of effort-based indicators and the need for process-aware assessment.

Keywords: Handwriting Trajectory · MLLM · Cognitive Patterns

1 Introduction

AI in education is shifting assessment from final outcomes to process-oriented analysis [14, 26]. However, most existing diagnostic and recommendation systems

Z. Sheng and S. Shen contributed equally to this work. Code and related materials: https://github.com/DobbySquirrel/Beyond_Correctness.

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still rely heavily on summative signals such as correctness and response time. These signals provide only a coarse view of learning and often lead to superficial personalization, such as mere repeated practice, without diagnosing how or why students' problem-solving processes break down [11]. At the same time, many process-oriented methods depend on intrusive or sparse data sources, such as think-aloud protocols and post-hoc questionnaires, which are difficult to deploy at scale and only weakly connected to real-time cognitive regulation [6].

Unlike such methods, handwriting trajectories captured by digital pens provide a non-intrusive yet promising source of naturally occurring process data from real-world exams. Generated unobtrusively, they avoid the recall bias of questionnaires [7] and the behavioral disruption of think-aloud protocols [10]. Temporal pauses, writing order, and revision behaviors provide rich signals of planning, execution, and self-regulation [9, 18]. Despite their richness, prior work has predominantly focused on kinematic features (e.g., velocity profiles, pressure dynamics) or employed machine learning models for classification [2, 17]. While predictive, these approaches often overlook the semantic context of the handwriting, limiting interpretability and generalization across tasks and learners. A fundamental challenge is that the same handwriting behavior can serve different cognitive functions depending on when and why it occurs. Rewriting symbols may reflect motor correction, conceptual restructuring, or verification. Crossing out expressions may signal exploration of alternatives or abandonment of a plan. Without modeling this context dependence, process data remain cognitively opaque, making it hard to derive reliable indicators of metacognitive monitoring and strategic regulation from handwriting alone.

Inspired by classical problem-solving and self-regulated learning theories [20, 22, 34], our framework treats problem-solving as a phased process, using stage information as a backbone to contextualize behaviors. Rather than using multimodal large language models (MLLMs) as black-box classifiers, we leverage them to operationalize these stages into an interpretable semantic mapping from trajectories to cognitive functions. This yields a structured behavioral representation that links raw (x, y, t) traces to process-level transitions, supporting multi-resolution analysis of micro-level regulatory events, question-level dynamics, and stable student-level profiles.

Using one semester of real-world exam data, we conduct a multi-level pattern analysis that yields three key insights corresponding to different scales of problem-solving. At the micro-level, corrections serve stage-dependent roles as repairs during execution and checks during verification (Sect. 4.1). At the student level, clustering reveals profiles like Reflective Builders and Immediate Responders (Sect. 4.2). At the performance level, a strategy-style mismatch reveals that process efficacy depends on cognitive style, where impulsive learners often exhibit inefficient diligence (Sect. 4.3).

These findings suggest that interpretation of process data requires considering not only what actions occur, but when and for whom they function as effective regulation, providing a theoretical foundation for stage-aware diagnostics and adaptive learning support. Overall, our main contributions are:

- **Authentic Trajectory Dataset.** A large-scale dataset (300k+ strokes) collected under authentic exam conditions, capturing ecologically valid process data to bridge the gap between lab studies and real-world diagnostics.
- **Stage-Aware Behavioral Representation.** A framework leveraging MLLMs to map raw traces into a semantic representation, enabling transparent, theory-grounded analysis of cognitive processes beyond black-box prediction.
- **Cognitive-Style-Dependent Insights.** Empirical evidence shows that cognitive style moderates the impact of effort, revealing profiles like inefficient diligence, where impulsive learners engage intensely yet perform poorly.

2 Related Work

2.1 LLMs in Automated Educational Assessment

Recent advances in Large Language Models (LLMs) have significantly expanded the scope of automated educational assessment, enabling richer representations of student behavior beyond traditional hand-crafted features [8, 23]. LLMs have been applied to evaluating complex tasks such as student explanations in programming [4] and collaborative learning dynamics through interaction modeling [21]. Beyond direct assessment, LLMs have been integrated into both intelligent tutoring systems, such as ChatTutor [5] which leverages chain-of-thought reasoning for sustained support, and student simulation frameworks [16] that characterize learner profiles. Despite these advances, most existing systems rely on linguistic artifacts (e.g., chat logs) or focus on grading final outcomes, overlooking the non-linguistic dynamics of self-regulation. To bridge this gap, we employ MLLMs to interpret handwriting trajectories, mapping raw strokes to cognitive stages.

2.2 Measuring Behavior via Handwriting Trajectories

Handwriting trajectories captured by digital pens offer granular signals for tracing cognitive processes [3, 17]. Prior studies have leveraged kinematic attributes and graphical production metrics to estimate cognitive load and engagement [13, 15, 31]. However, in open-ended tasks, these features often risk analytical tautology, as the volume of solution production inherently confounds with grading criteria. We address this by targeting multiple-choice questions, effectively decoupling regulatory dynamics from the correctness of the final answer.

Regarding the modeling of these trajectories, recent research has evolved from low-level stroke segmentation [32] toward process-oriented frameworks that reconstruct reasoning chains [29]. Yet, these approaches often rely on heuristics, lacking an intermediate semantic structure to contextually interpret behavior. To bridge this gap, our methodology is grounded in established problem-solving theories [12, 22]. By employing MLLMs to map raw signals onto structured cognitive stages, we move beyond descriptive traces to provide theory-grounded diagnostics, ensuring that extracted patterns are interpretable and actionable.

3 Dataset Curation

3.1 Data Collection and Preprocessing

Our dataset was collected from a semester of in-class examinations in a secondary school, following prior digital pen studies [29, 31]. The study involved 16 students (8 male, 8 female; aged 17–19) and was approved by the institutional review board. Students and their guardians provided informed consent. Data were collected in ordinary classroom exams using digital pens and coordinate-embedded answer sheets, recording pen-state, spatial coordinates, and timestamps. Across seven examinations in mathematics, physics, and chemistry, the dataset contains 70 student-exam instances, 379 unique questions, and 3,154 question attempts. Teacher-assigned question scores served as ground-truth performance labels. To protect privacy, the raw dataset is not publicly released.

Multi-scale Temporal Segmentation. At the most fundamental stage of processing, individual raw pen strokes are grouped into continuous writing segments, each corresponding to an uninterrupted pen-down burst. Consecutive trajectory segments that are separated by temporal gaps of less than 3s are subsequently consolidated into higher-level trajectory groups. We adopt this 3s threshold to rigorously distinguish between low-level motor latency and high-level cognitive planning. Interruptions below this duration typically reflect neuromuscular adjustments or transcription fluency constraints, whereas longer pauses are robust indicators of higher-level cognitive processes such as semantic retrieval and strategic restructuring [15, 19, 30, 32]. By filtering out these mechanical interruptions, we ensure that the resulting 12,587 trajectory groups represent coherent units of writing activity and distinct instances of local cognitive engagement.

Spatial Partitioning and Question Mapping. Trajectory groups are mapped to specific questions on the answer sheet via spatial partitioning, yielding 3,154 question attempts across the dataset. Since a single attempt typically comprises multiple disjoint trajectory groups, we aggregate them to extract question-level temporal features. Specifically, we compute total writing time (W) and total inter-segment pause time (P) to quantify the balance between active execution and cognitive processing. For a question composed of K segments, where segment k spans $[t_k^{\text{start}}, t_k^{\text{end}}]$, these metrics are defined as:

$$W = \sum_{k=1}^K (t_k^{\text{end}} - t_k^{\text{start}}), \quad P = \sum_{k=1}^{K-1} (t_{k+1}^{\text{start}} - t_k^{\text{end}}) \quad (1)$$

These units capture the internal rhythm of the problem-solving process, distinguishing between rapid procedural fluency and pause-heavy reasoning phases. These segment-level timings underpin the aggregates in Sect. 3.5 and the process-performance analyses in Sect. 4. To visualize the problem-solving process, we create progressive visualizations for each question. As shown in Fig. 1, we sequentially accumulate trajectory groups, with previous groups colored blue and the current group highlighted in green. This visualization captures writing dynamics, supporting the annotation process in Sect. 3.3.

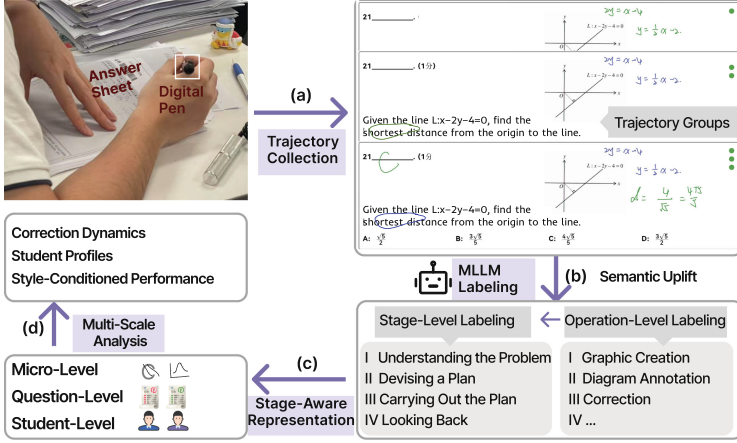


Fig. 1. Workflow of the stage-aware framework. (a) Segmentation of raw handwriting trajectories into discrete trajectory groups. (b) MLLM-driven mapping of these groups to functional operations situated within theory-grounded problem-solving stages. (c) Construction of a multi-level behavioral representation spanning micro, question, and student scales. (d) Downstream analysis revealing stage-dependent correction dynamics, student profiles, and style-conditioned performance.

3.2 Theoretical Taxonomy

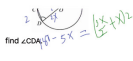
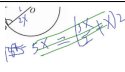
We introduce a hierarchical framework to transform raw trajectories into cognitive signals, organizing behavior into fine-grained operations and coarse-grained stages. Unlike traditional surface-level statistics [15, 30, 32], this multi-level integration resolves the cognitive ambiguity inherent in isolated actions.

Operation-Level Functional Units. The first layer characterizes the nature of student actions by abstracting raw spatiotemporal signals into discrete functional operations. As detailed in Table 1, we define a taxonomy of seven operation types organized into two broad categories. The first category, *Creation*, encompasses behaviors involved in generating new content, including Graphic Creation, Writing and Final Answer. The second category, *Annotation*, covers interactions with existing content, such as Problem Text Annotation, Diagram Annotation, Answer Marking, and Correction. This detailed labeling transforms physical pen strokes into a sequence of goal-oriented actions.

Stage-Level Cognitive Context. This dimension answers the question of when the action occurs within the problem-solving logic. Grounded in classical problem-solving theories [20, 22, 34], we segment the solution process into four canonical cognitive stages supplemented by a special case category. (1) *Understanding the Problem* involves information encoding, where students extract data and identify unknown variables from the problem statement. (2) *Devising a Plan* represents the strategic search phase, characterized by model construction, pattern recognition, or formula retrieval. (3) *Carrying Out the Plan* constitutes the algorithmic execution phase, focusing on symbolic manipulation and logical

derivation. (4) *Looking Back* functions as the metacognitive verification phase for examining the solution’s reasonableness. Additionally, we introduce a *Special Case* category to capture impulsive responses where students bypass these intermediate reasoning stages to provide an immediate answer.

Table 1. Taxonomy of Operation-Level Functional Units. The framework distinguishes between generative Creation and interactive Annotation groups, with definitions and visual examples for each operation.

Group	Operation	Definition	Example
Creation	Graphic Creation	Drawing new, independent visual representations in blank areas (e.g., diagrams, free-body sketches, coordinate axes).	
	Writing	Producing new textual, numerical, or symbolic content in blank areas (e.g., formulas, calculations, notes).	
	Final Answer	Writing or submitting the final result in a designated answer area (e.g., answer box).	25 
Annotation	Problem Text Annotation	Marking or highlighting the printed problem text (e.g., underlining keywords, circling given conditions).	Which is <u>incorrect</u> ?
	Diagram Annotation	Adding auxiliary geometric elements or labels onto printed diagrams or figures (e.g., auxiliary lines, angle or point labels).	
	Answer Marking	Indicating selection or elimination of predefined multiple-choice options (e.g., circling or crossing out A, B, C, D).	A. $\angle COB = \angle AOB$ 
	Correction	Revising or voiding previously written content (e.g., crossing out formulas, overwriting numbers).	

3.3 MLLM-Based Automated Annotation

Implementation Details and Minimal Reproducibility. We used Gemini-3-Pro as the multimodal annotator under a fixed prompt and deterministic decoding. Each call received (i) a crop of the current trajectory group, (ii) the accumulated history visualization, (iii) the predefined taxonomy, and (iv) a constrained instruction to return one label from a closed set with a brief rationale.

Operation-Level Labeling. To decode the functional intent of each trajectory group into the categories defined in Table 1, we implemented a context-aware prompting strategy. The MLLM is provided with a visual crop where the current trajectory is highlighted in green and historical traces in blue (see Fig. 1). To minimize the stochasticity of open-ended classification, we formulated the

labeling as a constrained reasoning task governed by a hierarchical decision logic. The prompt first distinguishes Creation from Annotation, then assigns one operation label from the predefined closed set based on spatial anchors and local visual context. This structured prompting ensures that functional intent is derived from logical elimination rather than simple pattern matching, thereby grounding classifications in observable visual evidence.

Stage-Level Labeling. To operationalize problem-solving stages, our framework achieves visual-temporal integration by representing the solution process as a sequential image strip (see Fig. 1(b)). This representation explicitly encodes the temporal evolution of the problem-solving state. By injecting temporal meta-data and the operation labels derived in the previous step into the multimodal context, we transform stage inference into a context-aware sequence labeling task. This enables the MLLM to synthesize functional intent with visual progression, decoding not just what was written, but when and how the cognitive state transitioned over time. By observing the logical flow, the model generates stage labels and reasoning traces that justify cognitive transitions. In practice, the stage-level prompt used the current group visualization, the accumulated trajectory history, and the previously assigned operation label as joint context, and required the model to assign one stage label from the predefined taxonomy.

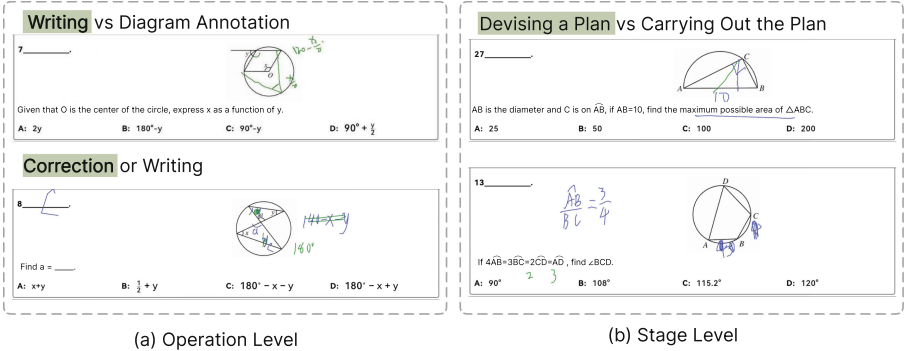


Fig. 2. Case study of robustness against semantic ambiguity. (a) the operation level (e.g., spatially distinguishing Diagram Annotation from Writing). (b) the stage level (e.g., identifying subtle transitions from Devising a Plan to execution).

3.4 Annotation Quality

Initial Annotation and Validation Protocol. The MLLM was initially provided with explicit category definitions for operations and stages (see Sect. 3.2). For validation, two expert annotators independently examined a stratified sample of 427 trajectory groups drawn from two complete examinations spanning multiple subjects and question types. Annotators were given the trajectory visualizations, the operation/stage definitions, and the original answer-sheet context,

and judged whether the MLLM-assigned labels were supported by the available visual evidence. Disagreements were resolved through discussion, and the resulting revision rules were incorporated back into the prompting protocol.

Robustness to Semantic Ambiguity. To validate the robustness of our framework, we conducted a targeted case study on boundary interactions, scenarios where visual cues are inherently noisy (see Fig. 2). We focused on two critical ambiguity types: (1) Spatial Ambiguity at the operation level, such as distinguishing Writing from Diagram Annotation when text visually encroaches on figures, or separating Correction from overwriting. (2) Sequential Ambiguity at the stage level, particularly the subtle transition between Devising a Plan and Carrying Out the Plan, where partial symbolic expressions could imply either strategy search or algorithmic execution. By incorporating the explicit disambiguation rules derived from these edge cases into the MLLM’s hierarchical logic, our framework effectively decouples these semantic overlaps that traditional heuristic methods often misclassify.

Final Agreement and Reliability. After iterative refinement, our automated annotation achieved high consistency with human judgment, recording 94.6% agreement (Cohen’s $\kappa \approx 0.92$) at the operation level and 88.5% ($\kappa \approx 0.84$) at the stage level. The high agreement confirms that the framework effectively resolves the semantic ambiguity of raw trajectories, successfully converting noisy physical signals into reliable cognitive evidence.

3.5 Multi-scale Feature Extraction

Based on the multi-dimensional behavioral representation, we derive multi-scale features to support analyses of regulation, strategy, and performance. These features are computed from verified annotations at three levels including micro-level events, question-level processes, and student level aggregates.

Micro-Level Features. We analyze the dynamics of correction events to decode the cognitive mechanics of real-time self-regulation. First, each correction bounding box B_c is spatially anchored to a target category c^* by maximizing cumulative overlap with prior trajectory groups:

$$c^* = \arg \max_c \sum_{g \in \mathcal{G}_c} \text{area}(B_c \cap B_g) \quad (2)$$

where $\mathcal{G}_c$ denotes the set of prior groups belonging to category c . A correction event is a trajectory group labeled as Correction. Its correction target c^* is the previously written category with the largest overlap with the correction ink, as defined in Eq. (2).

Beyond spatial association, we evaluate the cognitive cost of regulation through Regulatory Persistence. We identify the nearest preceding (j) and succeeding (k) creation-type groups relative to the correction and define the persistence indicator as $M = \mathbb{I}(\text{cat}(j) = \text{cat}(k))$. This metric distinguishes the nature of the intervention. A persistent state ($M = 1$) indicates a fluid local repair, where the student immediately resumes the prior activity with low cognitive

overhead (seamless transition). A mode switch ($M = 0$) signals a strategic pivot, implying that the error triggered a deeper re-evaluation of the problem-solving path (cognitive disruption).

Question-Level Features. At the question level, we summarize the temporal organization of the problem-solving process. Let $T(s, c)$ denote the total writing time spent in activity category c during stage s , where s corresponds to the four cognitive stages defined in Sect. 3.2. We derive two complementary stage-aware measures. The stage share ϕ_s represents the proportion of total writing time allocated to a specific phase s relative to the entire question attempt. The stage-internal proportion $\rho_{s,c}$ quantifies the functional emphasis on activity c within the total duration of stage s . These features characterize the cognitive weight of each stage and the distribution of activities within them.

Student-Level Features. Question-level metrics are aggregated into high-dimensional feature vectors to construct stable behavioral profiles across five behavioral dimensions (summarized in Table 2). These include session timing and activity volume, switching and diversity, correction intensity, as well as action and stage compositions. These features enable the identification of generalized learning styles from fine-grained handwriting data. Representative student-level features used for clustering are summarized in Table 2.

4 Analysis and Results

This section evaluates how the stage-aware framework transforms raw handwriting into interpretable cognitive signals by disambiguating micro-behaviors, identifying stable profiles, and examining style-conditioned effectiveness of effort.

4.1 Stage-Aware Interpretation of Micro-level Behaviors

We begin by examining whether incorporating problem-solving stages meaningfully changes the interpretation of low-level behaviors. A canonical example is correction behavior, which is typically treated as a uniform signal of error or revision in digital learning systems. Without contextual information, corrections are difficult to interpret. They may reflect careless mistakes, strategic refinement, or post-hoc verification. By situating correction events within cognitive stages, our framework resolves this ambiguity. For each correction, we assign a target category and a problem-solving stage as defined in Sect. 3.5. The target follows the overlap rule in Eq. (2), and the stage is that of the trajectory group containing the correcting ink. We analyze the $n=349$ events with both labels and report the corresponding targets and transitions below. Two consistent patterns emerge.

First, correction events exhibit strong behavioral continuity. In 68.5% of cases, students maintained Regulatory Persistence ($M = 1$), immediately resuming the prior activity category after a correction. This high degree of Regulatory Persistence indicates that most corrections function as local micro-repairs rather than disruptive strategy shifts. Second, correction targets differ

sharply across stages. Pooling the 349 events, a chi-square test of independence between problem-solving stage and correction target yields a strong association ($\chi^2 = 210.40$, $p < .001$, Cramér's $V = 0.448$). During the Carrying Out the Plan stage, 83.6% of corrections targeted Writing traces, consistent with real-time procedural monitoring and execution repair. In contrast, during the Looking Back stage, 53.1% of corrections targeted the Final Answer, reflecting solution verification and outcome checking.

Table 2. Representative student-level features used for clustering.

Aspect	Representative Features	Interpretation
Session timing and activity volume	Initial pause, gap ratio, total working time, number of action groups	Overall pace and amount of observable work across an exam session.
Switching and diversity	Category switch rate, stage switch rate, category entropy, stage entropy	How flexibly students move across action types and problem-solving stages.
Correction intensity	Correction count, corrections per minute	Frequency of explicit revision behavior.
Action composition	Time share and count for the 7 action categories; aggregate creation and annotation shares	Whether students rely more on content generation or interaction with existing content.
Stage composition	Time share and count for the 5 stages; Special Case rate	How effort is distributed across the stage-aware problem-solving process.

These findings demonstrate that stage awareness transforms correction from a generic mechanical act into a precise signal of self-regulation, effectively distinguishing execution-level maintenance from metacognitive validation. Further granularity is revealed when cross-referencing these dynamics with student profiles (see Sect. 4.2): while Reflective Builders maintained high Regulatory Persistence ($M = 1$) during execution, their infrequent mode switches ($M = 0$) predominantly occurred during the Looking Back stage. This suggests that for high-performing students, strategic pivots are not random disruptions but deliberate, stage-appropriate re-evaluations triggered during the verification phase.

4.2 Stage-Structured Processes and Student Behavioral Profiles

Having established the interpretability of stage-aware micro-level signals, we next examine whether these signals reflect stable individual differences rather than transient task effects.

We aggregated the question-level measures in Sect. 3.5 into 70 student-exam feature vectors for exploratory profiling. The clustering input comprised 70 student-exam instances and 37 standardized continuous features covering session timing, switching/dispersion, correction intensity, action composition, and stage composition (Table 2). Exam scores were excluded from clustering and used only for post-hoc profiling. We applied GMM and selected $K = 2$ by the highest silhouette score.

The first group, termed *Reflective Builders* ($N = 63$), achieved a high median score of 72.2. Their problem-solving behavior is characterized by structured progression across stages, moderate switching rates (median 0.52/min), and balanced time allocation between Carrying Out the Plan (41.5%) and Looking Back (35.9%). They also exhibited active self-monitoring, with a median of 4.0 corrections per exam. The second group, termed *Immediate Responders* ($N = 7$), achieved a substantially lower median score of 30.6. These students displayed behavioral stagnation, spending a median of 83.1% of their time in the Special Case stage, characterized by immediate answer attempts or unstructured pauses without intermediate reasoning. Their switching rates were negligible (median 0.03/min), and no correction behaviors were observed. The two groups showed differences across multiple process dimensions ($p < .05$).

External Validation. To confirm that these behaviorally derived profiles correspond to established cognitive constructs, we cross-referenced them with external cognitive style ratings. Specifically, two experienced teachers assessed each student using a five-point reflection-impulsivity scale. To ensure reliability, the teachers first performed independent assessments, achieving high inter-rater agreement (Cohen's $\kappa = 0.82$). The final scores were determined through a consensus process to resolve any discrepancies. Statistical comparison confirms a strong alignment. Reflective Builders exhibit significantly higher ratings than Immediate Responders (median = 3.0 vs. 2.0, $p < .001$). This external validation demonstrates that the stage-aware features successfully capture latent cognitive tendencies from unobtrusive behavioral data. While the Immediate Responders cluster is small, its behavioral signature is highly consistent across exams and externally validated by teacher ratings, suggesting a stable response style rather than a sampling artifact.

4.3 Style-Conditioned Effectiveness of Effort

While behavioral profiles explain how students approach problems, educational relevance ultimately depends on whether these behaviors translate into performance. We therefore examine how the relationship between stage-level effort and outcomes is moderated by cognitive style.

Results reveal that the effectiveness of effort is strongly contingent on how that effort is distributed across problem-solving stages. For the Carrying Out the Plan stage, we observe a significant negative interaction between reflection-impulsivity and execution effort ($\beta = -58.27$, $p < .01$), where β denotes the interaction coefficient. This indicates diminishing returns to extended execution for reflective students. For these learners, excessive procedural manipulation often signals strategic deadlock rather than progress. In contrast, impulsive students benefit significantly from execution effort, as it marks a constructive shift from guessing to structured calculation. An opposite moderation pattern emerges for the Special Case stage. The interaction between reflection-impulsivity and the weighted share of time spent in Special Case activities is positive and highly significant ($\beta = 48.38$, $p < .001$). While reflective students occasionally engage

in brief, unstructured activity as part of verification, impulsive students devote a large fraction of their time to this stage without corresponding gains.

These findings expose a phenomenon we term inefficient diligence. Observable effort, whether quantified as time spent or visible activity, does not guarantee productive problem-solving. When effort is misaligned with appropriate stages and regulatory behaviors, increased engagement may yield little benefit.

5 Discussion

This section discusses the theoretical and design implications.

5.1 Theoretical Implications

Validating Self-Regulated Learning. Conventional assessments often prioritize final results, obscuring the cognitive dynamics of the problem-solving process [14]. The identification of reflective builders and immediate responders (see Sect. 4.2) empirically validates classical theories of Self-Regulated Learning [34]. Our data supports this by showing a clear divergence in regulatory style: Reflective Builders utilize Regulatory Persistence ($M = 1$) to sustain procedural fluency during execution, reserving strategic pivots ($M = 0$) for the Looking Back stage. This targeted use of mode-switching reflects a sophisticated level of metacognitive control that immediate responders lack. By capturing these dynamics, our framework redefines the nature of student struggle: productive struggle is not merely the presence of errors, but the capacity for a strategic pivot when local repairs fail to resolve cognitive conflicts.

Cognitive Offloading and Externalization. Our findings highlight the role of cognitive offloading [33]. Reflective Builders treat the paper as an external cognitive buffer, offloading intermediate symbolic states via diagrams and annotations. This distributed cognitive strategy preserves precious central executive resources for high-level monitoring, effectively preventing the metabolic exhaustion of working memory seen in immediate responders. This aligns with prior research suggesting that handwriting is not merely a record of thought, but a constituent part of the thinking process itself that reduces cognitive load [13, 17]. The inefficient diligence we observed is essentially a failure of this externalization process, where students expend cognitive energy on retention rather than manipulation. The phenomenon of inefficient diligence suggests a strategic bottleneck: without the scaffold of externalization, immediate responders' efforts are trapped in a retention-heavy loop. Their extended execution duration ($\phi_{CarryingOut}$) represents a failure to prune incorrect solution paths early, resulting in a negative effort-to-outcome elasticity ($\beta = -58.27$) that outcome-based metrics completely fail to capture.

5.2 Design Implications

Real-time Process Diagnosis. Our identification of inefficient diligence provides a precise trigger for real-time intervention. Current systems typically intervene

based on error-count or latency. However, our findings suggest that for immediate responders, unstructured stagnation in the Special Case stage ($\phi_{SpecialCase}$) is a counterproductive signal. By monitoring Regulatory Persistence (M) and stage-switching rates, AI tutors can distinguish between productive persistence and unstructured stagnation. For instance, when the framework detects a high-frequency Special Case loop with a switch rate below $0.05/min$, the system could prompt the student to articulate a plan rather than provide another procedural hint. This shifts the adaptive logic from task-level support to process-oriented interventions [5], helping students transition from impulsive to reflective problem-solving archetypes.

Targeting Behavioral Weaknesses. The stable behavioral profiles suggest that problem recommendation should prioritize regulatory remediation over knowledge acquisition [6]. For immediate responders who habitually bypass planning phases, systems should curate tasks imposing structural constraints on reasoning. For instance, geometry proofs can mandate diagram labeling as a prerequisite for equation entry, effectively forcing the cognitive offloading that these students habitually bypass. This transforms practice into a behavior-shaping mechanism, exercising under-developed self-regulation through task-embedded scaffolds.

Explainable Human-AI Collaboration. This framework enhances Explainable AI (XAI) by converting noisy, high-frequency pen strokes into a transparent cognitive trajectory. While end-to-end models can grade answers effectively [4], they often fail to justify the reasons behind a student's struggle [11]. Our approach instead acts as a semantic lens for instructors. This enables a symbiotic relationship between AI and teachers [25,27]. The AI handles the high-frequency pattern detection of micro-behaviors, while the teacher interprets these patterns to provide high-level mentorship. Aggregated dashboards could identify students trapped in inefficient diligence, empowering teachers to intervene precisely where the cognitive breakdown occurs [28].

5.3 Limitations and Future Work

This study has several limitations. First, the dataset remains modest in scale, with repeated observations from the same students. Future work should validate these behavioral profiles on larger and more diverse cohorts. Second, although handwriting trajectories offer a rich window into cognition, they remain an indirect proxy. Future research should adopt a multimodal approach, integrating eye-tracking to capture internal looking-back behaviors or neural signals to validate the cognitive load associated with different stages [24]. Finally, our current annotation relies on the Gemini-3-Pro [1] in a post-hoc manner. Future work will focus on translating these insights into real-time interventions within authentic classroom settings.

6 Conclusion

This study introduces an interpretable framework that transforms raw handwriting trajectories into stage-aware cognitive signals. Our findings show that the

effectiveness of learning behaviors depends strongly on their temporal context. By shifting attention from written content to the regulation of thought, this work lays the groundwork for AI tutors that support cognitive processes rather than merely evaluate performance.

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